**4** **(a)** State **two** conditions required for two-source interference pattern to be observed.

1. .....

.....

2. .....

.....

[2]

**(b)** A structural engineer is using a laser-based monitoring system to detect microscopic shifts in a bridge's support beam.

A Young's Double Slit experiment where a monochromatic laser using red light of wavelength 633 nm was shone through two narrow slits attached to the beam. The resulting interference pattern is projected onto a screen located 2.50 m away.

At the start of the day, the distance between the centre of the central maximum and the third order maximum is measured to be 1.85 cm.

**(i)** Determine the separation between the two slits.

separation = ..... mm [3]

**(ii)** Later in the afternoon, due to thermal expansion, the slits physically move slightly further apart.

State and explain any changes to the **distance between the bright fringes** on the screen.

.....

.....

.....

..... [2]

**(iii)** As the light travels from the slits to the screen, it passes through regions of varying temperatures, which affects the refractive index of the air.

The refractive index $n$ of air is given by the expression

$$n = \frac{c}{v}$$

where $c$ is the speed of light in vacuum and $v$ the speed of light in air.

Suggest how the fluctuation in $n$ affects the interference pattern formed on the screen.

.....

.....

.....

..... [2]

**(c)** Khongoryn Els of the Gobi Desert is known as the Singing Dunes.

At the Khongoryn Els, a specific frequency is produced when a surface layer of dry, "boiling" sand slides over a more compact, stationary layer during an “avalanche”. This gap between the moving surface and the stationary bed (the vibrating layer) acts like a pipe **closed at one end** (the bottom) and **open at the top**. A 3-D schematic diagram of the layers is shown in Figure 4.1

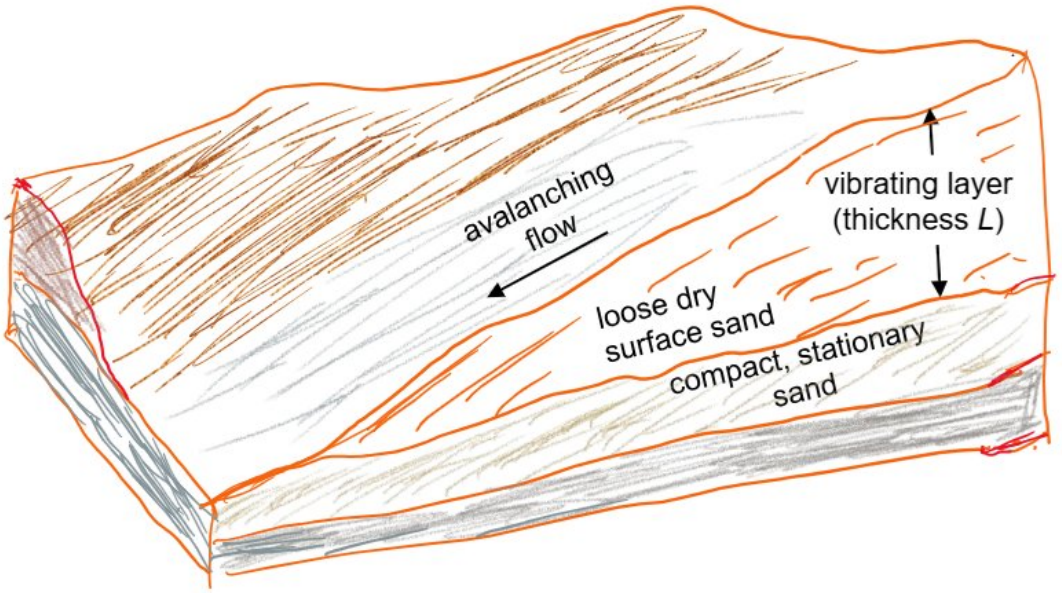



**Figure 4.1**

The thickness of this resonating dry sand layer $L$ is 1.20 m and the speed of sound in the hot, sandy air is 348 m $s^{-1}$.

Examiner's Use

**(i)** The vibrating dry sand layer of the singing dunes can be represented as a pipe of length 1.20 m as shown in Figure 4.2.

When the fundamental frequency is heard within this dry sand layer, on Figure 4.2,

**1.** label the positions of the pressure antinode **PA** and displacement antinode **DA**,

**2.** sketch the pressure variation of the standing wave pattern formed.

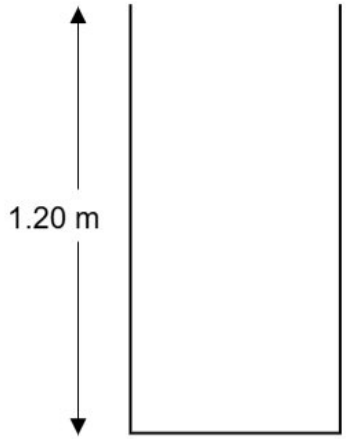



**Figure 4.2**

[2]

**(ii)** Calculate the fundamental frequency (the "hum") produced by the dune.

frequency = ..... Hz [1]

Examiner's Use

**(iii)** In addition to the fundamental frequency, the dune also produces higher-frequency overtones that contribute to its "eerie" sound.

Calculate the frequency of the next higher harmonic that could be produced within this 1.20 m sand layer.

frequency = ..... Hz [2]

[Total: 14]